Team Excercises, Chapter 1, Matrix Inverses

Matrix multiplication is weird, as we've noted, but it's also really important!

1. Try These!

(a) Calculate the product

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

(b) Calculate the product

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) Given an $m \times n$ matrix A, describe a matrix I where $IA = A$ What size is I? What are its entries?

(d) Given an $m \times n$ matrix A, describe a matrix J where $AJ = A$ What size is J? What are its entries?

(e) Complete the definition for the $n \times n$ **identity matrix**:

$$[I_n]_{ij} = \begin{cases} \quad & \text{if } i = j \\ \\ \quad & \text{if } i \neq j \end{cases}$$

(f) ***Turn-In! (Writing Assignment)***

Prove that I_n is the unique $n \times n$ identity matrix.

2. Reflect. For a square matrix A, what does it mean to say B is the multiplicative inverse of A?

If A has a multiplicative inverse, do you think it will be unique?

Example 1 Pause: let's talk about inverses and how to find them! □

Definition 2 Singular Matrices. A square matrix A is called **singular** if it does not have a (multiplicative) inverse. $\diamondsuit$

Definition 3 Special Matrices. A matrix $A = [a_{ij}] \in M_{n\times n}$ is called:

upper triangular	if all its entries below the main diagonal are 0 (that is, if $a_{ij} = 0$ whenever $i > j$);
lower triangular	if all its entries above the main diagonal are 0 (that is, if $a_{ij} = 0$ whenever $i < j$);
diagonal	if all its entries off the main diagonal are 0 (that is, if $i \neq j$ then $a_{ij} = 0$);
scalar	if it is equal to a constant times the identity matrix (that is, if there is a real number r for which $a_{ij} = r$ if $i = j$ and $a_{ij} = 0$ if $i \neq j$);
symmetric	if $A^T = A$;
skew symmetric	if $A^T = -A$.

$\diamondsuit$

3. Classificiation! For each given matrix, check the property (or properties) it has. For those rows where columns are checked, come up with a matrix that satisfies those definitions.

Table 4 Check the definitions that apply.

Matrix	Upper Tri.	Lower Tri.	Diagonal	Scalar	Symmetric	Skew Sym.
$\begin{bmatrix} 0 & 5 & 1 \\ 0 & 6 & 2 \\ 0 & 0 & 3 \end{bmatrix}$						
$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{bmatrix}$						
$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$						
$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 8 \end{bmatrix}$						
$\begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$						
$\begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$						
$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$						
$\begin{bmatrix} 5 & 4 & 0 \\ 4 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}$						

4. Conjectures. Brainstorm about the ways different definitions fit together. For example, are there matrices that are both symmetric and skew symmetric square matrix?

Remark 5 Writing Assignment Problems. Please start working on writing up the ***turn in*** question, and also see if you and your team can figure out how to ***prove that matrix multiplication is associative***. First start with the general 2×2 case, then the 3×3 case, then see if you can recognize where you are using properties of summations in the general case.

Properties of summation notation.

$$\sum_{i=1}^{n}(r_i + s_i)a_i = \sum_{i=1}^{n} r_i a_i + \sum_{i=1}^{n} s_i a_i$$

$$\sum_{i=1}^{n} c(r_i a_i) = c\left(\sum_{i=1}^{n} r_i a_i\right)$$

$$\sum_{i=1}^{n}\left(\sum_{j=1}^{m} a_{ij}\right) = \sum_{j=1}^{m}\left(\sum_{i=1}^{n} a_{ij}\right)$$

Finally, ***prove one of your conjectures***!